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(D) Newtons Third Law of Motion

3.13 What the Newton's first law tells us is that:

> To every action there is an equal and opposite reaction.

Ex:-
In order to jump, you apply a (force of action) on the ground and then as a result the ground exerts a (force of reaction) on you, therefore pushing you up.

So to what the Newtons third Law states in detail is that:

> In an interaction of two bodies A and B, the magnitude of reaction force by the body B on body B, is equal in magnitude to the action force, but their directions are opposite.

Exercise 3 D

2) To every action, there is an equal and opposite reaction.

4)
(a) The force applied on the bullet is the action and the recoil is the reaction

(b) The force applied on the nail is the action, the recoil faced by hammer is the reaction.

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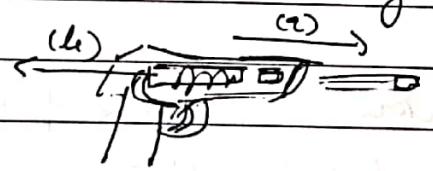
(c) The weight of the book is the action and the support force is the reaction force

(d) The action is the force powered by combustion and the reaction is the air molecules pushing the rocket forward.

(e) The force ~~exerted~~ exerted by the leg is the action force and the reaction is the ground pushing back on us, enabling us to walk.

(f) The action is the force exerted by the moving train and the reaction force is stationary train exerting a force back on the moving train.

6) When the hammer pushes the bullet with an action force called (a), the bullet also pushes back on the gun with reaction force (b).



7) So, to step ashore, you apply a force on the ~~[illegible]~~ boat ~~to~~ to be able to ~~[illegible]~~ step, the force pushes you to the shore, but if the boat is not tied, then the force you applied on the boat causes the boat to accelerate.

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MCQ's

1) (c) 2) (b)

Numerically

1) The wall exerts a force of 10 N in the west.
2) (a) 15 N down (weight caused by gravity)
(b) 15 N up (tension)

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